ETH zürich

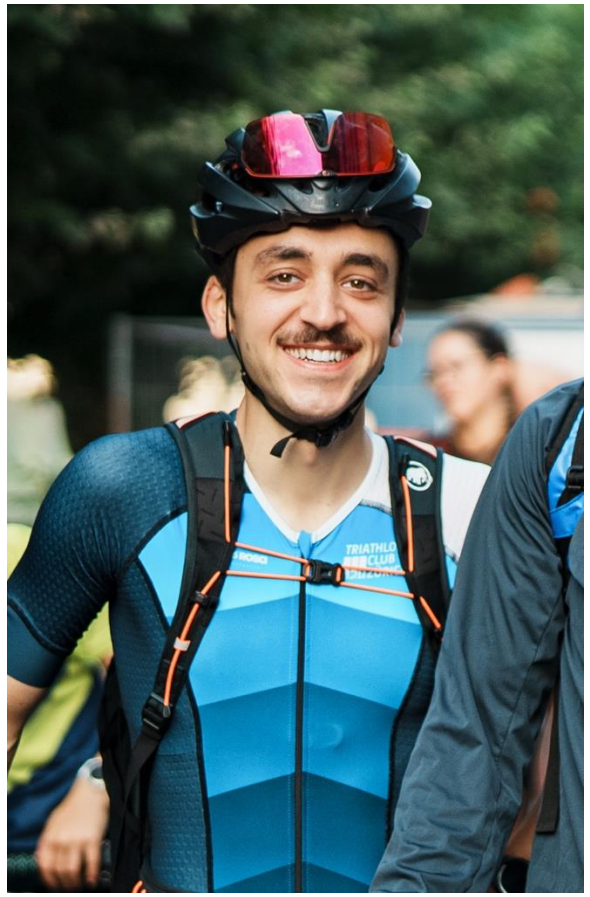
Department of Computer Science

Tamarin Workshop
Automated Protocol Verification

Felix Linker
PhD Student, ETH Zurich

Who am I?

- Felix Linker
  - Postdoc at ETH Zurich and independent consultant
  - Active at the IETF
- Worked on and with Tamarin for the last five years
  - Formal verification of the iMessage PQ3 protocol
  - Formal verification of the SecureDrop whistleblowing protocol
  - Implemented new induction scheme for Tamarin

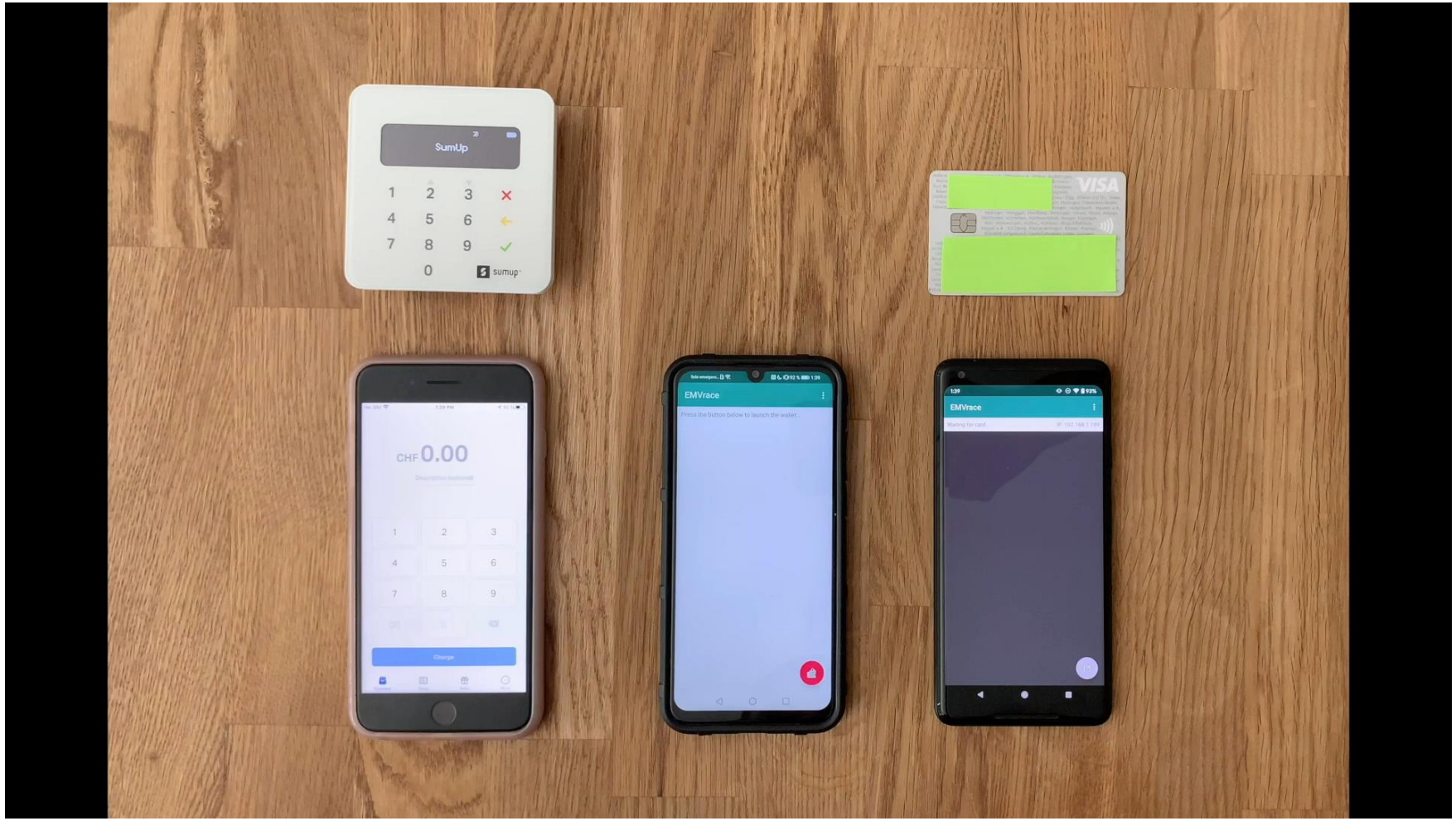


Part 1: An Introduction to Tamarin

The EMV Standard: Break, Fix, Verify

- S&P21 paper showed how to:
  - Pay with stolen credit card
  - Without ever needing the PIN

Attack Video



The EMV Standard: Break, Fix, Verify

- S&P21 paper showed how to:
  - Pay with stolen credit card
  - Without ever needing the PIN
- How did they find this attack?
- Used Tamarin!

The EMV Standard: Break, Fix, Verify

David Basin, Ralf Sasse, and Jorge Toro-Pozo
Department of Computer Science
ETH Zurich, Switzerland

IEEE COMPUTER SOCIETY

IEEE

42nd IEEE Symposium on Security and Privacy

Best Practical Paper

Sponsored by Intel and IBM

Presented to

David Basin, Ralf Sasse,
Jorge Toro-Pozo

for

The EMV Standard: Break, Fix, Verify

May 24-27, 2021

Alina Oprea, Program Chair SP 21 | Alvaro Cardenas, General Chair SP 21 | Thorsten Holz, Program Chair SP 21

What is Tamarin?

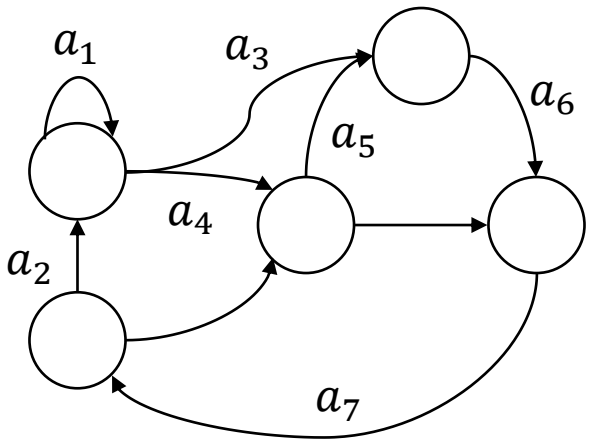
- Our world is powered by security-critical protocols
  - You want certain things to not happen
    - *Your neighbor reads your WhatsApp messages*
  - You want certain things to happen
    - *When you receive a message from me, it was I who sent it to you*
- Protocols are complex!
- People make mistakes!

With Tamarin, you can prove that a protocol (model) provides security properties

The Tamarin Prover

Multiset-Rewriting Rules…

- Define labelled state transition system
- Model participant steps
- Model environment



An equational Theory…

- Defines cryptographic operations
- Defines adversary capabilities

```
verify(sign(m, sk), m, pk(sk)) = true

sdec(senc(m, k), k) = m
```

Multiset Rewriting Rules

- You write a protocol **model**
- The model admits a set of **traces**
- A **multiset-rewriting rule** defines a state transition
  - Multiset-rewriting rules use multisets of **facts** (linear or **persistent**)
  - Pre-condition: Which facts are required to apply the rule?
    - Linear facts: Single-use
    - Persistent facts: Multiple uses
  - Labels: For reference in properties
  - Post-condition: Which facts are added when the rule is applied?

Labels

Post-condition

Pre-condition

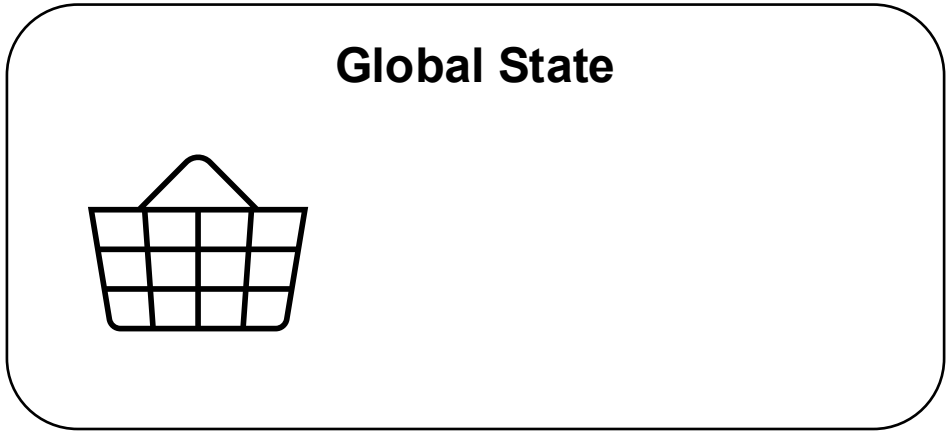
```
rule NAME:
  [ ... ] --[ ... ]-> [ ... ]
```

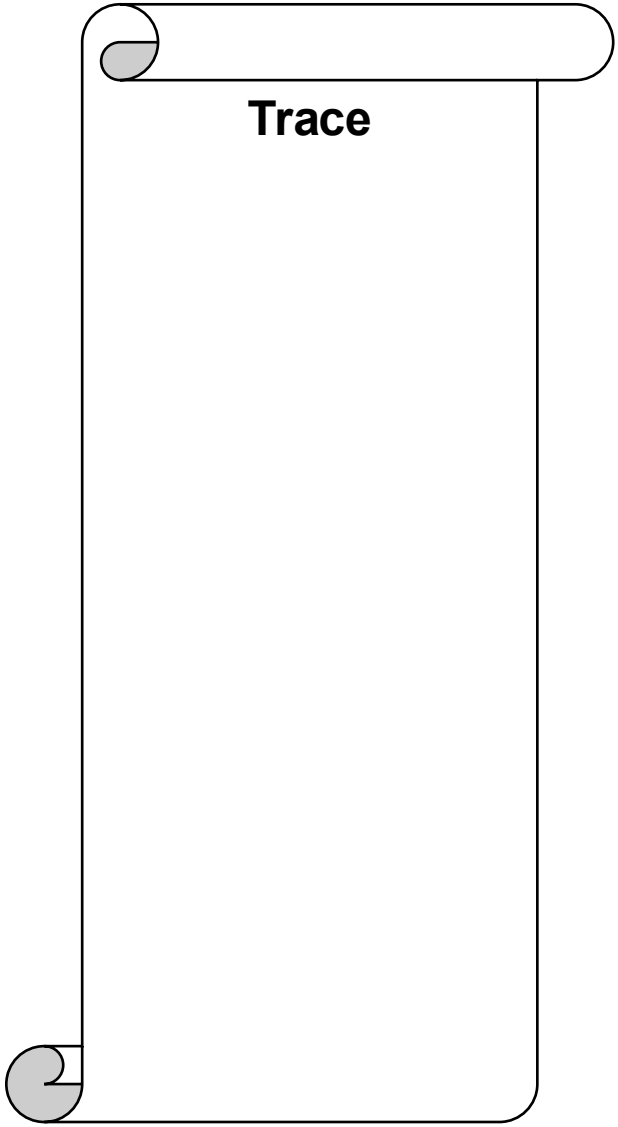
Multiset Rewriting Rules - Example

```
rule Start:
  [] --[ Start() ]-> [ Green(), Blue() ]
rule AddInfiniteRed:
  [ Green(), Blue() ] --[ RedAdded() ]-> [ !InfiniteRed() ]
rule AddYellow:
  [ Blue(), !InfiniteRed() ] --[ YellowAdded() ]-> [ Yellow() ]
```

Multiset Rewriting Rules - Example

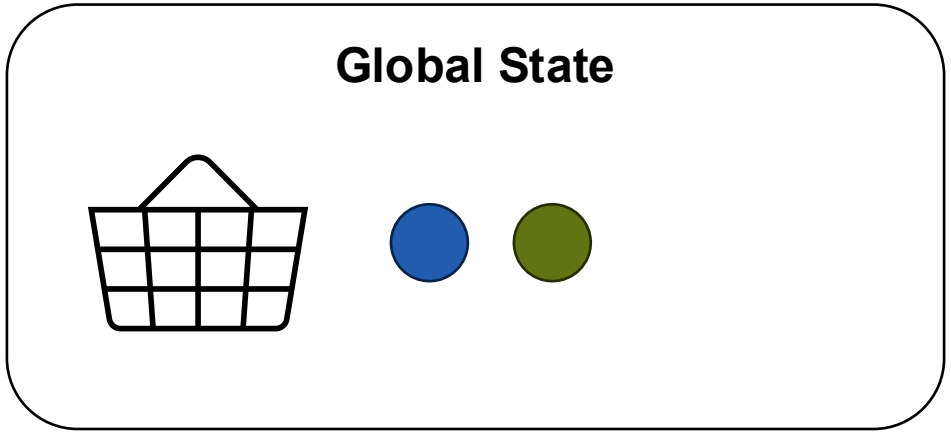
```
rule Start:
  [] --[ Start() ]-> [ Green(), Blue() ]
rule AddInfiniteRed:
  [ Green(), Blue() ] --[ RedAdded() ]-> [ !InfiniteRed() ]
rule AddYellow:
  [ Blue(), !InfiniteRed() ] --[ YellowAdded() ]-> [ Yellow() ]
```

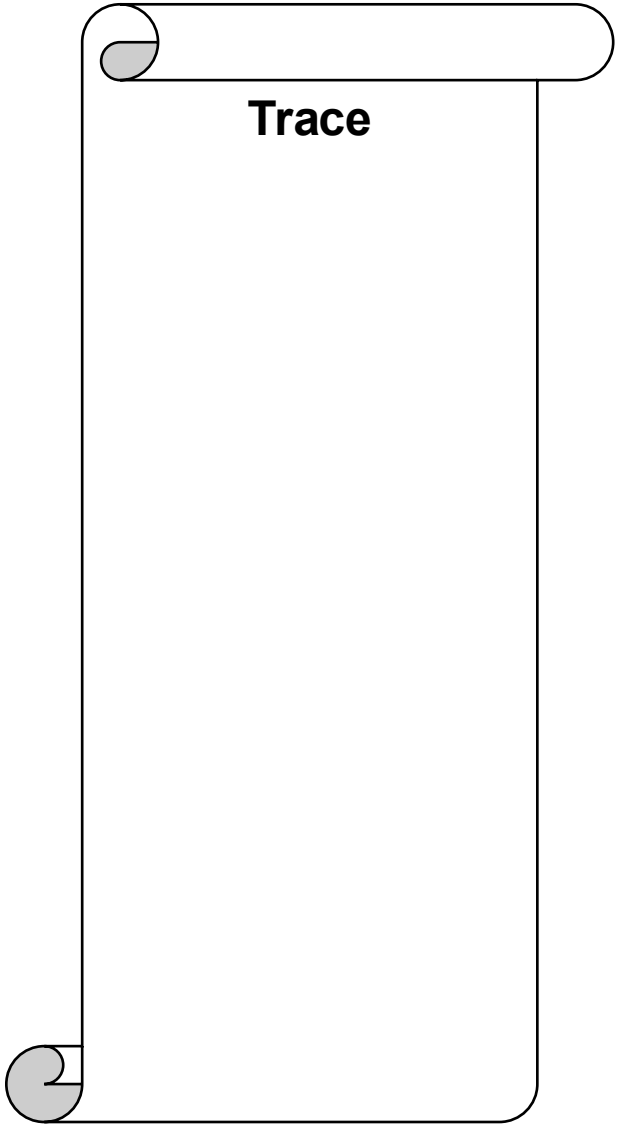




Multiset Rewriting Rules - Example

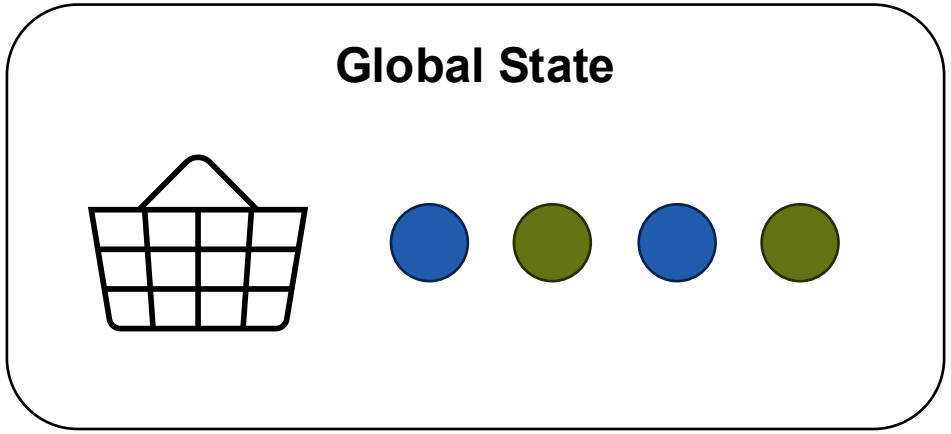
```
rule Start:
  [] --[ Start() ]-> [ Green(), Blue() ]
rule AddInfiniteRed:
  [ Green(), Blue() ] --[ RedAdded() ]-> [ !InfiniteRed() ]
rule AddYellow:
  [ Blue(), !InfiniteRed() ] --[ YellowAdded() ]-> [ Yellow() ]
```

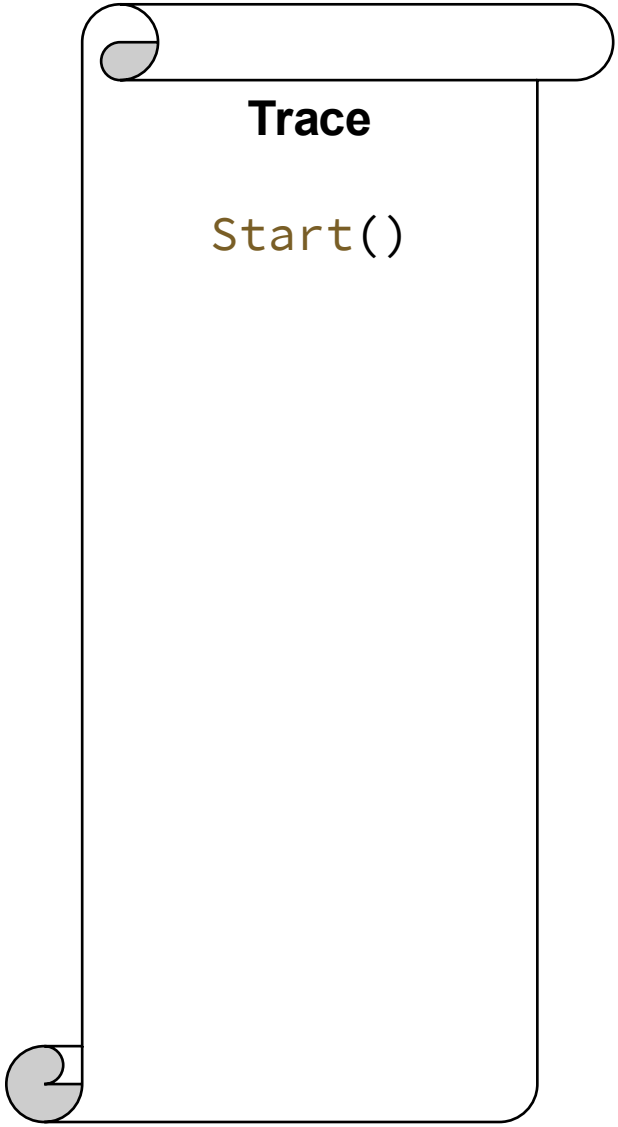




Multiset Rewriting Rules - Example

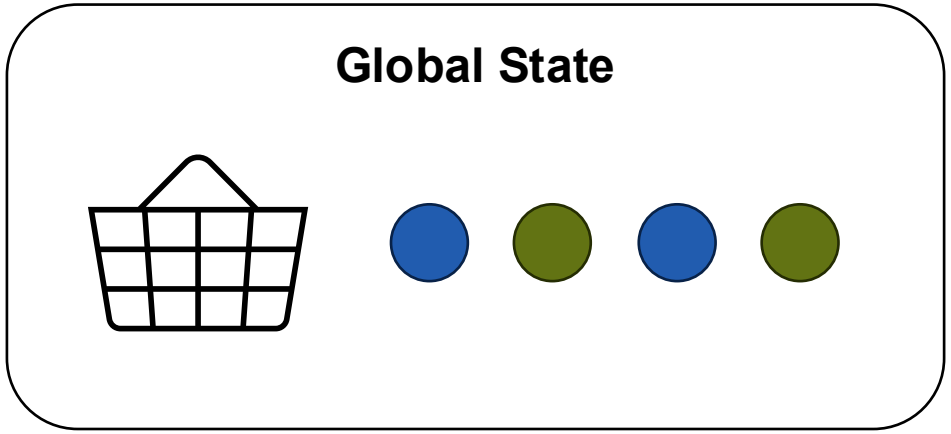
```
rule Start:
  [] --[ Start() ]-> [ Green(), Blue() ]
rule AddInfiniteRed:
  [ Green(), Blue() ] --[ RedAdded() ]-> [ !InfiniteRed() ]
rule AddYellow:
  [ Blue(), !InfiniteRed() ] --[ YellowAdded() ]-> [ Yellow() ]
```

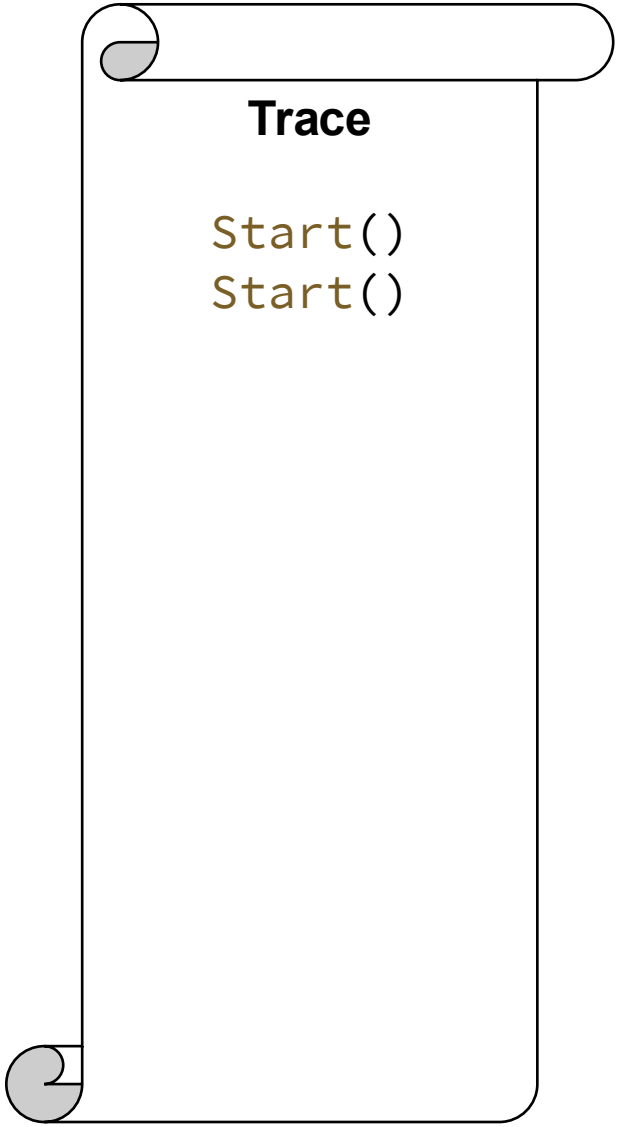




Multiset Rewriting Rules - Example

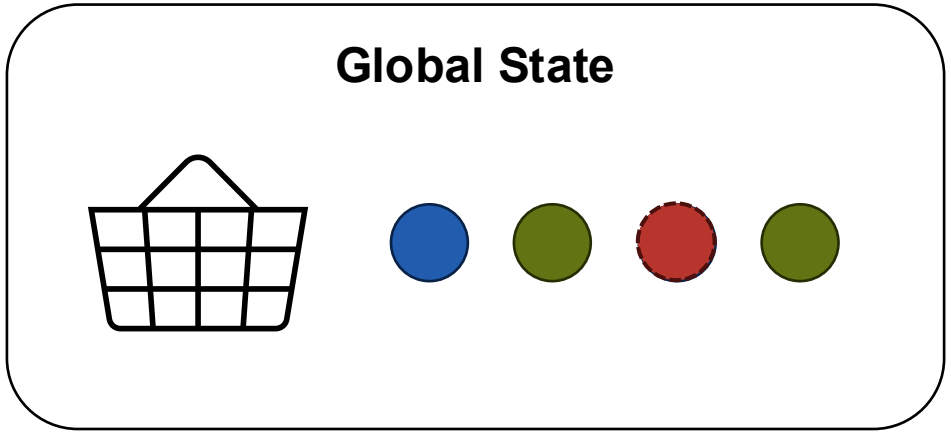
```
rule Start:
  [] --[ Start() ]-> [ Green(), Blue() ]
rule AddInfiniteRed:
  [ Green(), Blue() ] --[ RedAdded() ]-> [ !InfiniteRed() ]
rule AddYellow:
  [ Blue(), !InfiniteRed() ] --[ YellowAdded() ]-> [ Yellow() ]
```

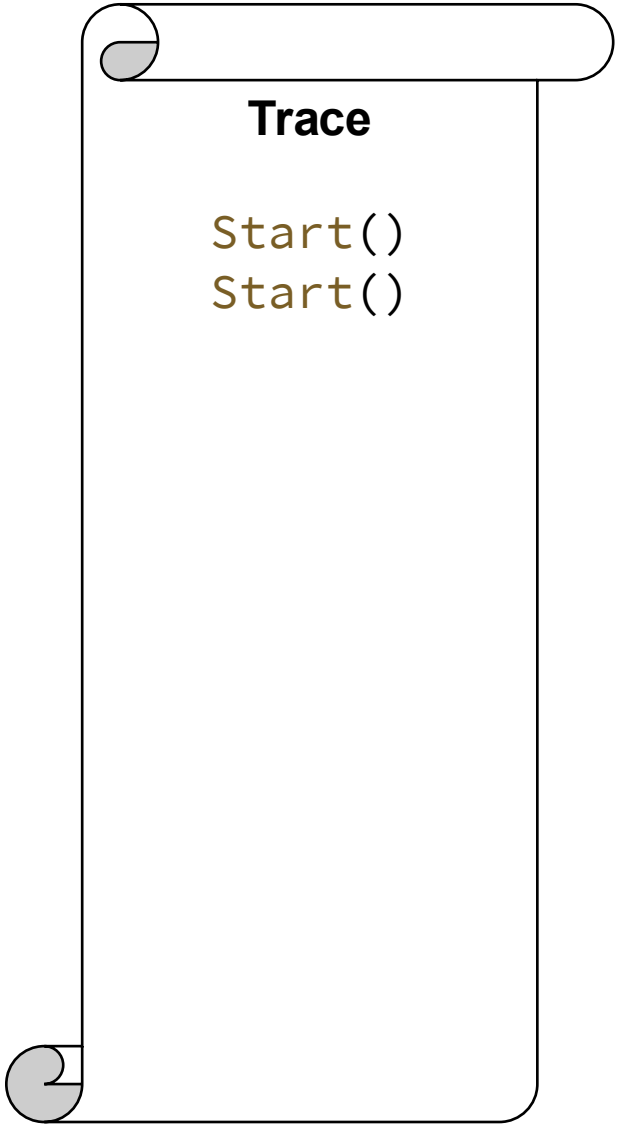




Multiset Rewriting Rules - Example

```
rule Start:
  [] --[ Start() ]-> [ Green(), Blue() ]
rule AddInfiniteRed:
  [ Green(), Blue() ] --[ RedAdded() ]-> [ !InfiniteRed() ]
rule AddYellow:
  [ Blue(), !InfiniteRed() ] --[ YellowAdded() ]-> [ Yellow() ]
```





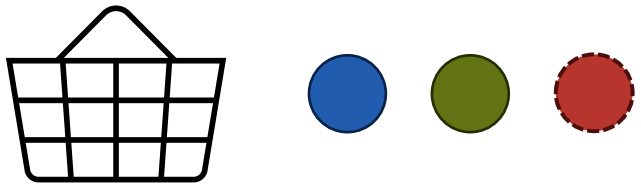
Multiset Rewriting Rules - Example

```
rule Start:
  [] --[ Start() ]-> [ Green(), Blue() ]
rule AddInfiniteRed:
  [ Green(), Blue() ] --[ RedAdded() ]-> [ !InfiniteRed() ]
rule AddYellow:
  [ Blue(), !InfiniteRed() ] --[ YellowAdded() ]-> [ Yellow() ]
```

Trace

```
Start()
Start()
RedAdded()
```

Global State



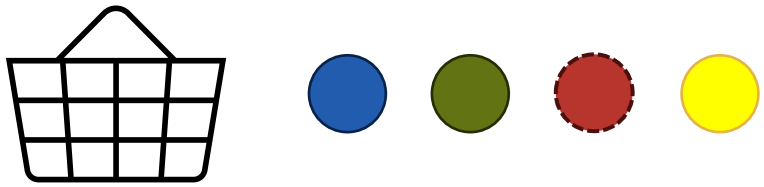
Multiset Rewriting Rules - Example

```
rule Start:
  [] --[ Start() ]-> [ Green(), Blue() ]
rule AddInfiniteRed:
  [ Green(), Blue() ] --[ RedAdded() ]-> [ !InfiniteRed() ]
rule AddYellow:
  [ Blue(), !InfiniteRed() ] --[ YellowAdded() ]-> [ Yellow() ]
```

Trace

```
Start()
Start()
RedAdded()
```

Global State



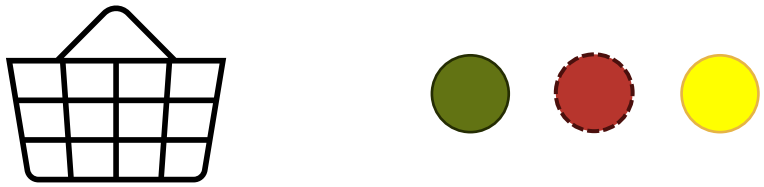
Multiset Rewriting Rules - Example

```
rule Start:
  [] --[ Start() ]-> [ Green(), Blue() ]
rule AddInfiniteRed:
  [ Green(), Blue() ] --[ RedAdded() ]-> [ !InfiniteRed() ]
rule AddYellow:
  [ Blue(), !InfiniteRed() ] --[ YellowAdded() ]-> [ Yellow() ]
```

Trace

```
Start()
Start()
RedAdded()
YellowAdded()
```

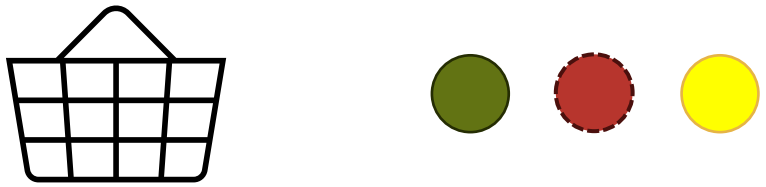
Global State



Multiset Rewriting Rules - Example

```
rule Start:
  [] --[ Start() ]-> [ Green(), Blue() ]
rule AddInfiniteRed:
  [ Green(), Blue() ] --[                                  ]
rule AddYellow:
  [ Blue(), !InfiniteRed                                  ow() ]
```

Demo



Properties talk about **traces**

Trace

```
Start()
Start()
RedAdded()
YellowAdded()
```

Multiset Rewriting Rules

- You write a protocol **model**
- The model admits a set of **traces**
- A **multiset-rewriting rule** defines a state transition
  - Multiset-rewriting rules use multisets of **facts** (linear or **persistent**)
  - Pre-condition: Which facts are required to apply the rule?
    - Linear facts: Single-use
    - Persistent facts: Multiple uses
  - Labels: For reference in properties
  - Post-condition: Which facts are added when the rule is applied?

Labels

Post-condition

Pre-condition

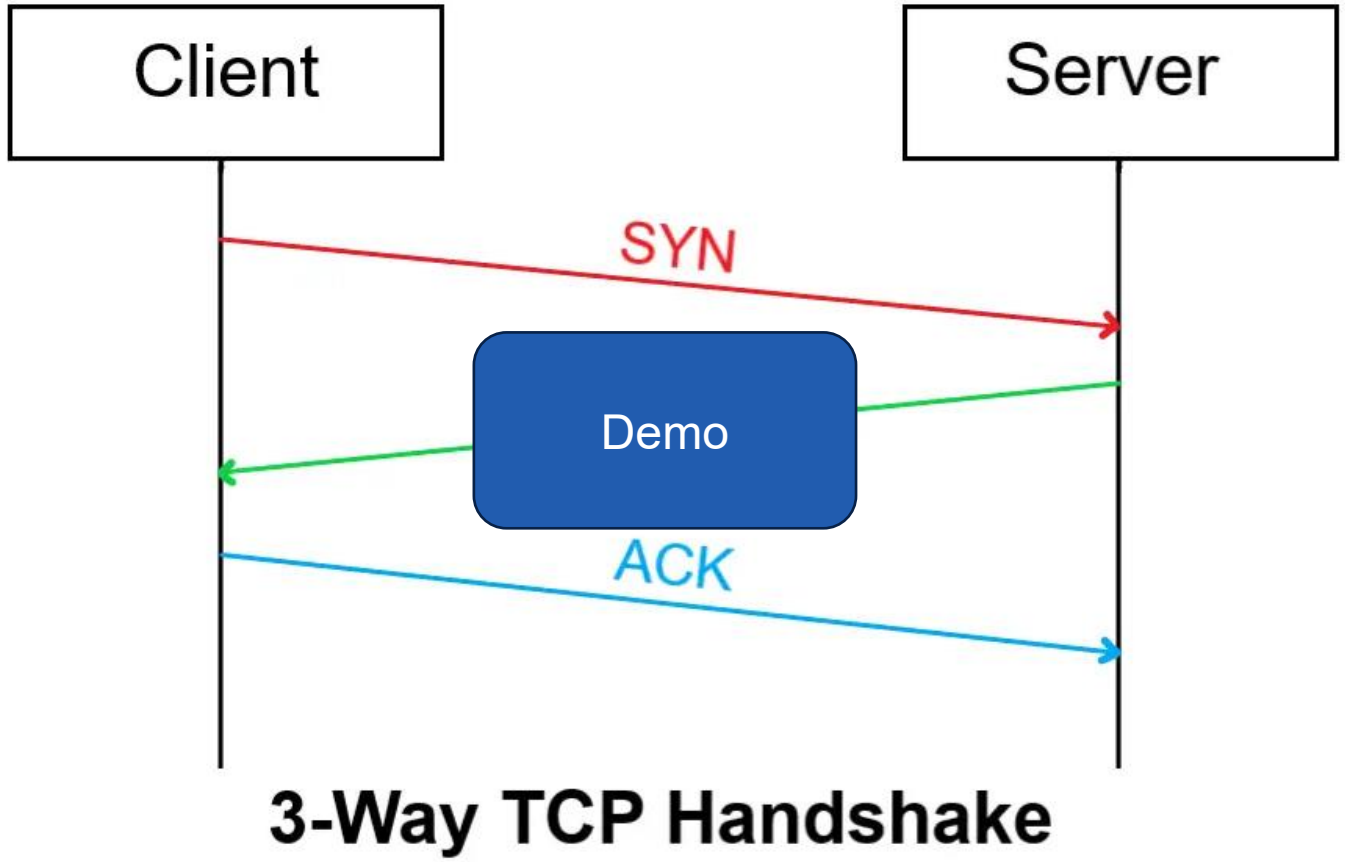
```
rule NAME:
  [ ... ] --[ ... ]-> [ ... ]
```

Values in Tamarin

- Facts can have **parameters**, which are **terms**
  - `Out(x)`
- Terms can be:
  - Constants: `'g'`
  - Unguessable (fresh) values: `~k`
  - Public values: `$P`
  - Function application: `f(t1, t2)`
- A variable `x` can be any term (also called **message**)
- **Equational theory** defines semantics of functions

```
functions: sign/2, verify/3, pk/1, true/0
equations: verify(sign(m, sk), m, pk(sk)) = true
```

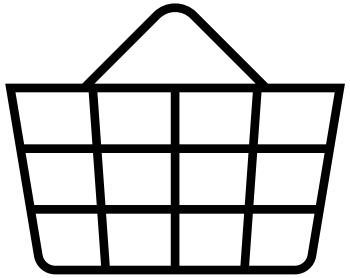
Example: TCP



Example: TCP – What happens under the hood?

```
rule SYN:
  []
  --[ Begin() ]->
  [ St_AliceWait(), Out('SYN’) ]
```

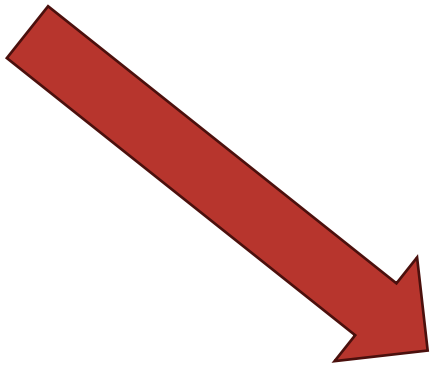
```
rule SYNACK:
  [ In('SYN') ]
  -->
  [ St_BobWait(), Out('SYNACK') ]
```

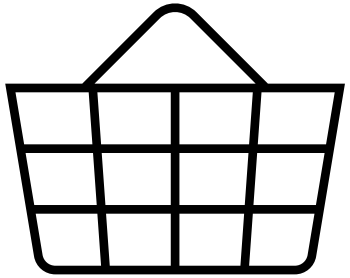


Example: TCP – What happens under the hood?

```
rule SYN:
  []
  --[ Begin() ]->
  [ St_AliceWait(), Out('SYN’) ]
```

```
rule SYNACK:
  [ In('SYN') ]
  -->
  [ St_BobWait(), Out('SYNACK') ]
```





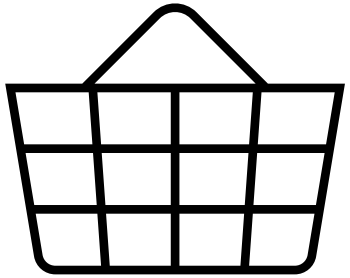
```
St_AliceWait()

Out('SYN')
```

Example: TCP – What happens under the hood?

```
rule SYN:
  []
  --[ Begin() ]->
  [ St_AliceWait(), Out('SYN’) ]
```

```
rule SYNACK:
  [ In('SYN') ]
  -->
  [ St_BobWait(), Out('SYNACK') ]
```



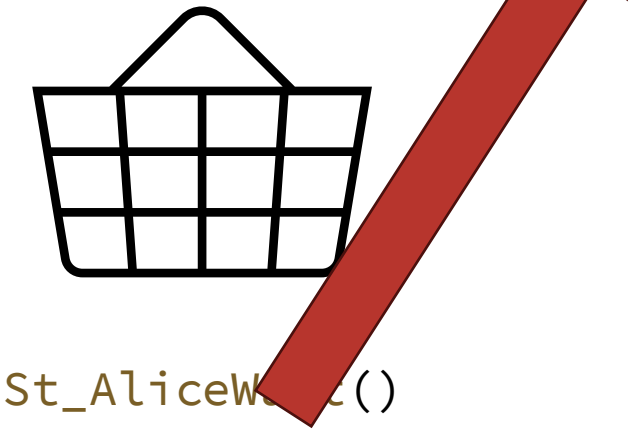
```
St_AliceWait()

In('SYN')
```

Example: TCP – What happens under the hood?

```
rule SYN:
  []
  --[ Begin() ]->
  [ St_AliceWait(), Out('SYN’) ]
```

```
rule SYNACK:
  [ In('SYN') ]
  -->
  [ St_BobWait(), Out('SYNACK') ]
```



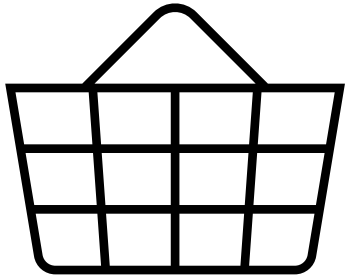
St_AliceWait()

In('SYN')

Example: TCP – What happens under the hood?

```
rule SYN:
  []
  --[ Begin() ]->
  [ St_AliceWait(), Out('SYN’) ]
```

```
rule SYNACK:
  [ In('SYN') ]
  -->
  [ St_BobWait(), Out('SYNACK') ]
```



St_AliceWait()

Example: TCP – What happens under the hood?

```
rule SYN:
  []
  --[ Begin() ]->
  [ St_AliceWait(), Out('SYN') ]
```

```
rule SYNACK:
  [ In('SYN') ]
  -->
  [ St_BobWait(), Out('SYNACK') ]
```

```
St_AliceW
```

```
St_BobWait()      Out('SYNACK')
```

Recap and Exercises

Message in

State write

State read

```
rule Standard:
  [ St_X0(...), In(term1) ] --[ Begin() ]-> [ St_X1(...), Out(term2) ]

functions: function/1

rule Memorize:
  [] --> [ Fact(function('x')) ]

rule LookUpAndSend:
  [ Fact(v) ] --> [ Out(v) ]
```

Message out

Unification

Recap and Exercises

1. Go to github.com/felixlinker/tamarin-workshop/
2. Clone or download
3. Install Tamarin
4. Do exercises 1+2

Hints

- Repository contains syntax cheat sheet
- If Tamarin warns you about one of the following, ask for help:
  - “Guardedness”
  - “Partial deconstructions”

Summary – Part 1

- So far you learned
  - Modelling in Tamarin
  - State-read/message-in + state-write/message-out pattern
  - The symbolic model
- Interested in more?
  - Documentation: https://tamarin-prover.com/manual/
  - Book: https://tamarin-prover.com/book/
- Also:
  - Manual proofs
  - Custom proof heuristics
  - Induction

Part 2: Analyzing Specifications with Tamarin

What is Tamarin?

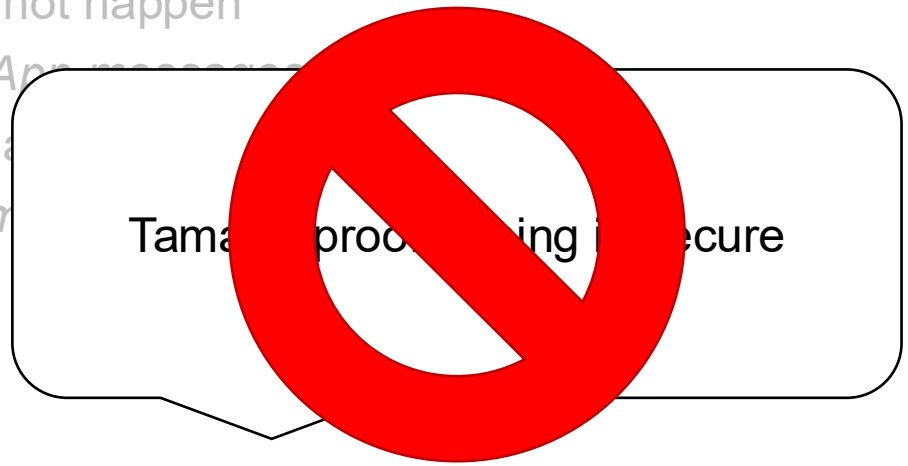
- Our world is powered by security-critical protocols
  - You want certain things to not happen
    - *NSA reads your WhatsApp messages*
  - You want certain things to a
    - *Merchant receives paym*
- Protocols are complex!
- People make mistakes!

Tamarin proof = thing is secure

With Tamarin, you can prove that a protocol (model) provides security properties

What is Tamarin?

- Our world is powered by security-critical protocols
  - You want certain things to not happen
    - *NSA reads your WhatsAp*
  - You want certain things to a
    - *Merchant receives paym*
- Protocols are complex!
- People make mistakes!



With Tamarin, you can prove that a protocol (model) provides security properties

What is Tamarin?

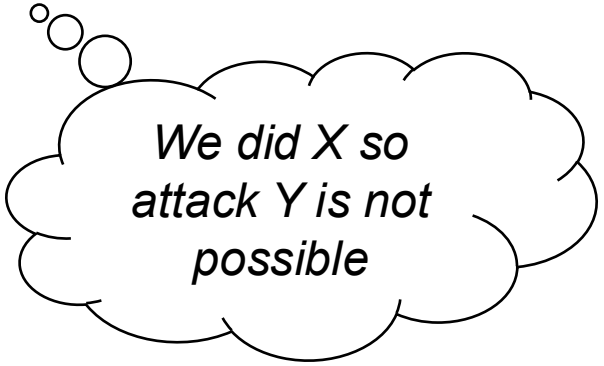
- Our world is powered by security-critical protocols
  - You want certain things to not happen
    - *NSA reads your WhatsApp messages*
  - You want certain things to always happen
    - *Merchant receives payment upon confirmation*
- Protocols are complex!
- People make mistakes!

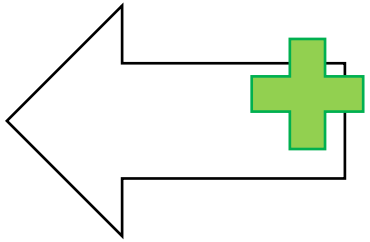
With Tamarin, you can prove that a protocol (model) provides certain security properties under certain assumptions

Specifications vs Formal Analysis

Specification

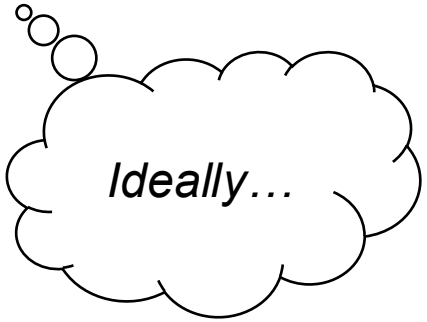
- Designed to foster compatible implementations
- Often deliberately underspecified
- Security considerations often ad-hoc





Formal Analysis

- A structured way to approach security
  - A positive definition of security properties
  - A list of <u>explicit</u> assumptions



Case Study: OAuth 2.0

10. Security Considerations 53
10.1. Client Authentication 53
10.2. Client Impersonation 54
10.3. Access Tokens 55
10.4. Refresh Tokens 55
10.5. Authorization Codes 56
10.6. Authorization Code Redirection URI Manipulation 56
10.7. Resource Owner Password Credentials 57
10.8. Request Confidentiality 58
10.9. Ensuring Endpoint Authenticity 58
10.10. Credentials-Guessing Attacks 58
10.11. Phishing Attacks 58
10.12. Cross-Site Request Forgery 59
10.13. Clickjacking 60
10.14. Code Injection and Input Validation 60
10.15. Open Redirectors 60
10.16. Misuse of Access Token to Impersonate Resource Owner in Implicit Flow 61

Case Study: OAuth 2.0 – Prior Work

Home > Conferences > CCS > Proceedings > CCS '16 > A Comprehensive Formal Security Analysis of OAuth 2.0

RESEARCH-ARTICLE

A Comprehensive Formal Security Analysis of OAuth 2.0

Authors: Daniel Fett, Ralf Küsters, Guido Schmitz Authors Info & Claims

CCS '16: Proceedings of the 2016 ACM SIGSAC Conference on Computer and Communications Security • October 2016 • Pages 1204–1215 • https://doi.org/10.1145/2976749.2978385

Published: 24 October 2016 Publication History

Check for updates

94 2,166

Get Access

CCS '16: Proceedings of the 2016 ACM SIGSAC... A Comprehensive Formal Security... Pages 1204–1215

← Previous Next →

ABSTRACT

The OAuth 2.0 protocol is one of the most widely deployed authorization/single sign-on (SSO) protocols and also serves as the foundation for the new SSO standard OpenID Connect. Despite the popularity of OAuth, so far analysis efforts were mostly targeted at finding bugs in specific implementations and were based on formal models which abstract from many web features or did not provide a formal

Fett, Küsters, Schmitz. CCS'16.

```
Workgroup: Web Authorization Protocol                    T. Lodderstedt
Internet-Draft:                                                  SPRIND
draft-ietf-oauth-security-topics-24                         J. Bradley
Updates: 6749, 6750, 6819 (if approved)                         Yubico
Published: 23 October 2023                                  A. Labunets
Intended Status: Best Current Practice            Independent Researcher
Expires: 25 April 2024                                        D. Fett
                                                             Authlete
```

OAuth 2.0 Security Best Current Practice

Abstract

This document describes best current security practice for OAuth 2.0. It updates and extends the OAuth 2.0 Security Threat Model to incorporate practical experiences gathered since OAuth 2.0 was published and covers new threats relevant due to the broader application of OAuth 2.0.

Discussion Venues

This note is to be removed before publishing as an RFC.

Discussion of this document takes place on the Web Authorization Protocol Working Group mailing list (oauth@ietf.org), which is archived at https://mailarchive.ietf.org/arch/browse/oauth/.

Source for this draft and an issue tracker can be found at https://github.com/oauthstuff/draft-ietf-oauth-security-topics.

Status of This Memo

This Internet-Draft is submitted in full conformance with the provisions of BCP 78 and BCP 79.

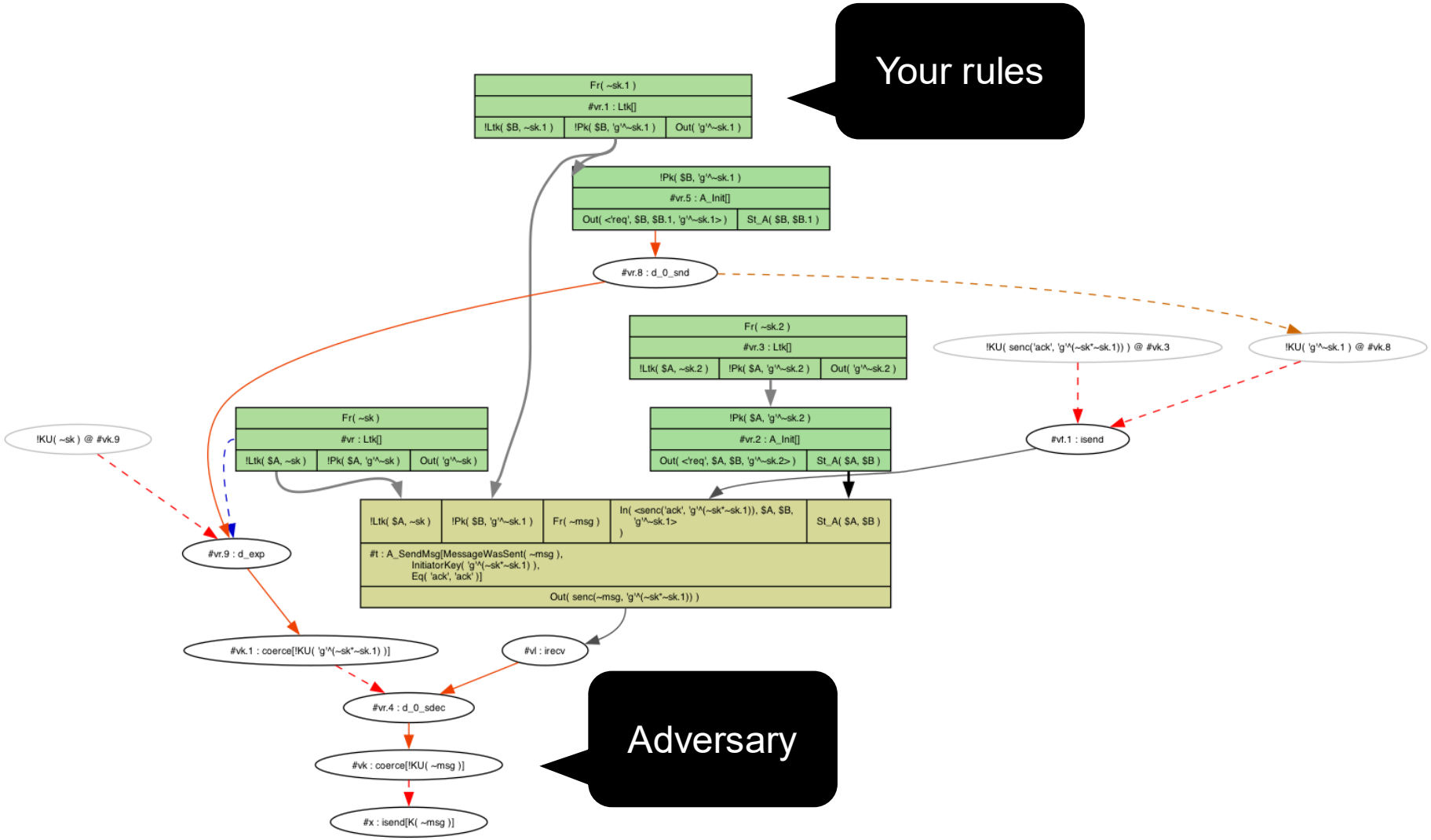
- But: Also doesn't list desired properties

Case Study: OAuth 2.0 – But how analyze a specification?

1. Implement an initial specification
2. Model security properties
   - It's okay if they are trivially true
3. Make your model more realistic
   - Now the properties are hopefully false
4. Refine everything
   - Let your understanding guide you
   - Let Tamarin tell you why your understanding is wrong

Use the GUI

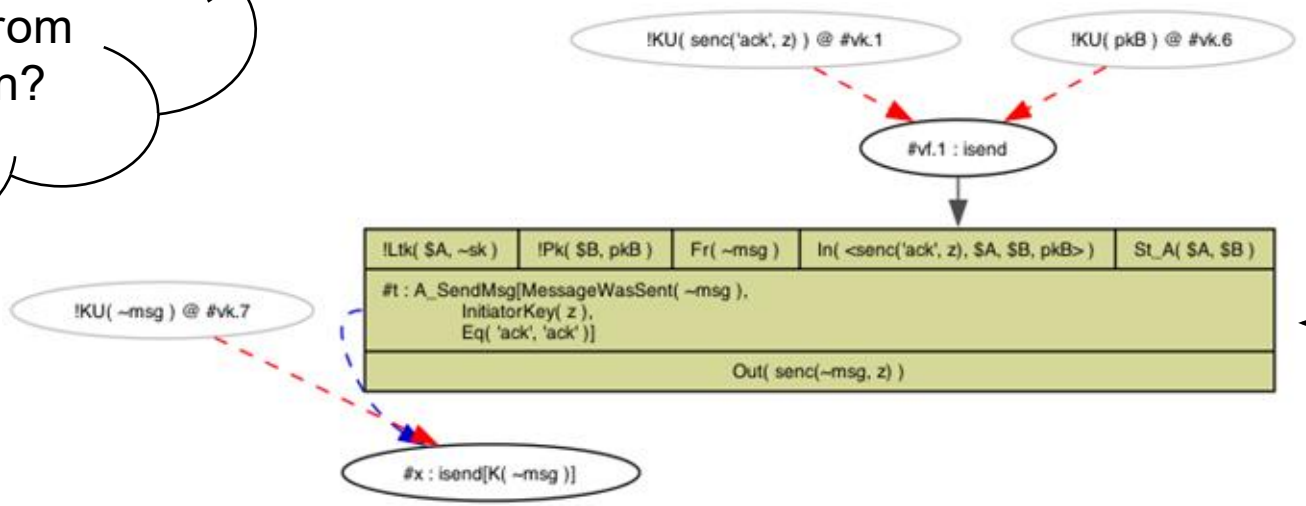
But how analyze a specification?



But how analyze a specification?

```
lemma SecrecyMessage:
  "All m #t. MessageWasSent(m) @ #t
   ==> not Ex #x. K(m) @ #x"
```

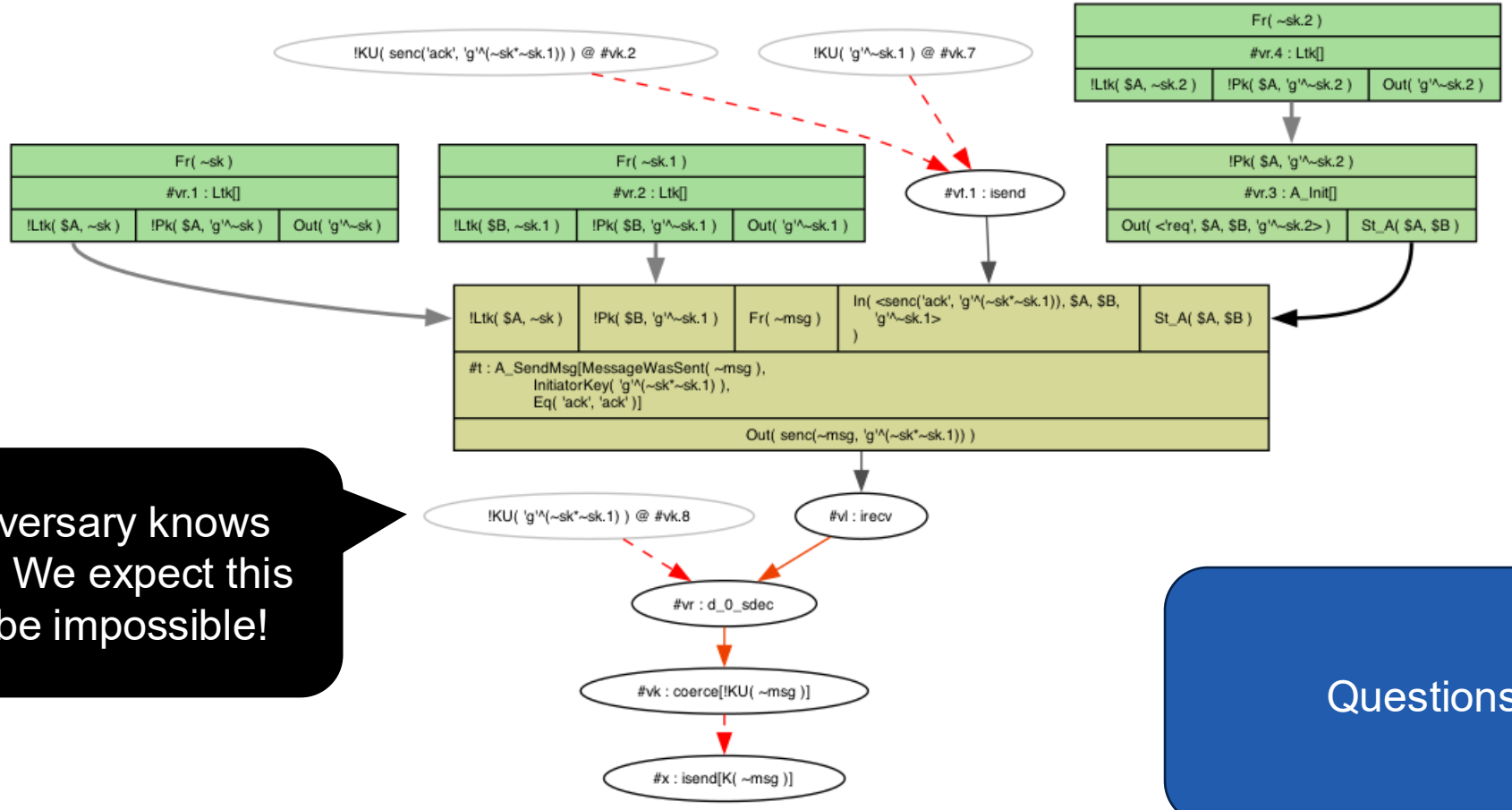
Where does the model deviate from our expectation?



Message was sent…

…and adversary knows it

But how analyze a specification?

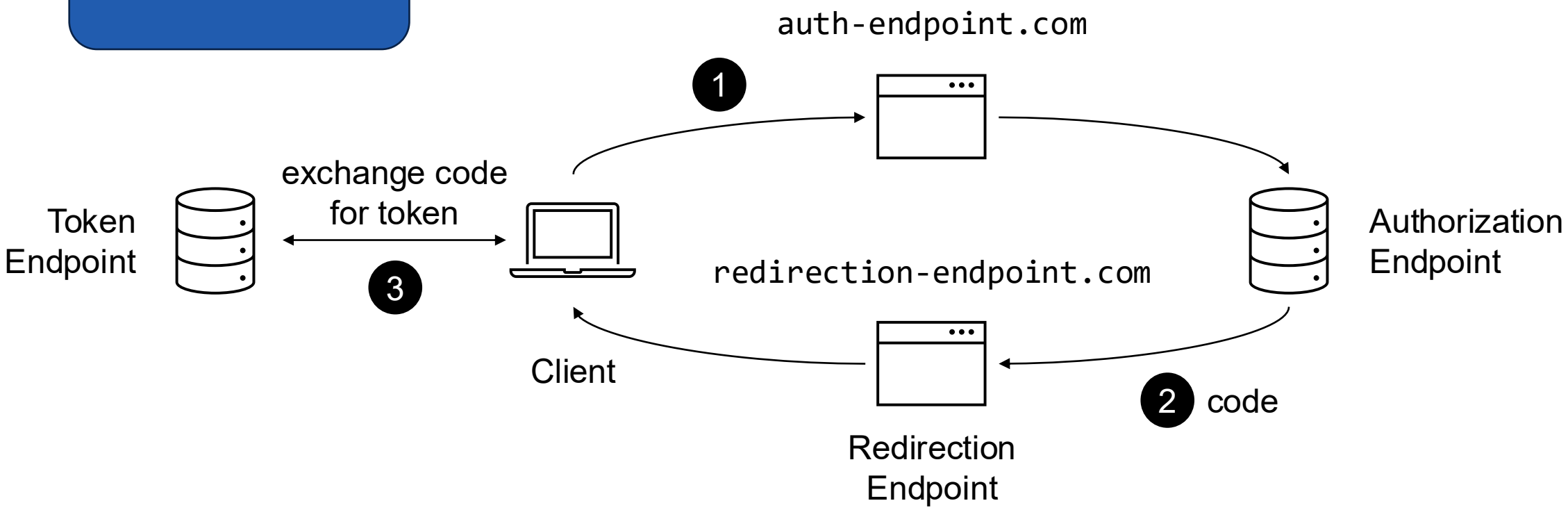


Adversary knows key! We expect this to be impossible!

Questions?

Case Study: OAuth 2.0 – Authorization Code Flow

Let's do it!



Further Reading

C. Herley and P. C. Van Oorschot, "SoK: Science, Security and the Elusive Goal of Security as a Scientific Pursuit," 2017 IEEE Symposium on Security and Privacy (SP), San Jose, CA, USA, 2017, pp. 99-120, doi: 10.1109/SP.2017.38.

Daniel Fett, Ralf Küsters, and Guido Schmitz. 2016. A Comprehensive Formal Security Analysis of OAuth 2.0. In Proceedings of the 2016 ACM SIGSAC Conference on Computer and Communications Security (CCS '16). Association for Computing Machinery, New York, NY, USA, 1204–1215. https://doi.org/10.1145/2976749.2978385